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Section: F

(Q:1)

$$Ax - \lambda Ix = 0$$

$$(A - \lambda I)x = 0 \rightarrow (i)$$

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

$$\lambda = \pm i$$

$$\lambda_0 = -i \text{ or } \lambda_1 = i$$

By using (i) $\lambda_1 = i$

$$\left(\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} \right) x = 0$$

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\left[\begin{array}{cc|c} -i & -1 & 0 \\ 1 & -i & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -i & 0 \\ -i & -1 & 0 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & -i & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$x_1 = ix_2$$

$$x_2 = x_2$$

$$x = \begin{bmatrix} i \\ 1 \end{bmatrix} x_2$$

$$E_i : \text{Span} \left\{ \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$$

$$\text{By using (i)} \quad \lambda = -i$$

$$\begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\left[\begin{array}{cc|c} i & -1 & 0 \\ 1 & i & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & i & 0 \\ i & -1 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{cc|c} 1 & i & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$x = \begin{bmatrix} -i \\ 1 \end{bmatrix} x_2$$

$$E_{-i} = \text{Span} \left\{ \begin{bmatrix} -i \\ 1 \end{bmatrix} \right\}$$

Basis exist but not real, basis will be complex. So, complex eigebasis exist.

(Q: 2)

$$(A - \lambda I)x = 0 \rightarrow (i)$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 \\ 0 & -1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(-1-\lambda) = 0$$

$$(-1)(+1+\lambda)(1-\lambda) = 0$$

$$1^2 - \lambda^2 = 0$$

$$1 = \lambda^2$$

$$\lambda = \pm 1$$

for $\lambda = 1$

$$\begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & -2 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 0 & -2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$x_1 = x_1$$

$$x_2 = 0$$

$$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} x_1$$

for $\lambda = -1$

$$\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\left[\begin{array}{cc|c} 2 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$x_1 = 0$$

$$x_2 = x_2$$

$$x = \begin{bmatrix} 0 \\ 1 \end{bmatrix} x_2$$

$$E_1 : \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

$$E_{-1} : \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

(Q:3)

if $\lambda = 0$, then;

$$(A - 0I)x = 0$$

$$Ax = 0 \rightarrow (i)$$

$$|A - \lambda I| = 0$$

$$|A| = 0 \rightarrow (ii)$$

Since $|A| = 0$, A will have infinite solutions as A is singular. Therefore, $\text{NUL}(A) \neq \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ or any Dimension} \right\}$ in

any case. Also since matrix is singular, its inverse does not exist and it is not invertible.

(Q:4)

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - 8 = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$\lambda = \frac{4 \pm \sqrt{16 + 20}}{2}$$

$$\lambda = 5 \text{ or } \lambda = -1$$

for $\lambda = 5$

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -4 & 2 & | & 0 \\ 4 & -2 & | & 0 \end{bmatrix} \sim \begin{bmatrix} -4 & 2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1/2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$x_1 = \frac{1}{2}x_2$$

$$x_2 = x_2$$

$$x = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} x_2$$

$$E_5 = \text{Span} \left\{ \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} \right\}$$

for $\lambda = -1$

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 2 & | & 0 \\ 4 & 4 & | & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$x_1 = -1x_2$$

$$x_2 = x_2$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} x_2$$

$$E_{-1} = \text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

Both of these represents lines passing through the origin in the direction of vector.

(Q:5)

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 & 1 \\ 0 & -\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$-\lambda(1-\lambda)^2 = 0$$

$$-\lambda(1+\lambda^2-2\lambda) = 0$$

$$-\lambda - \lambda^3 + 2\lambda^2 = 0$$

$$\lambda^3 - 2\lambda^2 + \lambda = 0$$

$$\lambda(\lambda^2 - 2\lambda + 1) = 0$$

$$\lambda = 0 \quad \text{or} \quad \lambda = 1$$

for $\lambda = 0$

for $\lambda = 1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 = -x_2$$

$$x_2 = x_2$$

$$x_3 = 0$$

$$x = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} x_2$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\left[\begin{array}{ccc|c} 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_1$$

$$E_0 = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$E_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$